

ROS2 ATOSFleetManagement Cheatsheet

Supported native installations

Ubuntu 20.04 -> ROS 2 Foxy
Ubuntu 22.04 -> ROS 2 Humble
Ubuntu 24.04 -> ROS 2 Jazzy

Always source ROS before building or launching

Ubuntu 24.04: source /opt/ros/jazzy/setup.bash
Ubuntu 22.04: source /opt/ros/humble/setup.bash
Ubuntu 20.04: source /opt/ros/foxy/setup.bash
Check: echo \$ROS_DISTRO

Local workspace build (manual)

```
cd <ATOS_WORKSPACE>
source /opt/ros/<jazzy|humble|foxy>/setup.bash
colcon build --symlink-install --cmake-args -DWITH_TRUCK_OBJECT_CONTROL=ON
source install/setup.bash
```

FleetManagement helper build

```
cd <ATOS_REPO_ROOT>
source /opt/ros/<jazzy|humble|foxy>/setup.bash
./scripts/build_atosfleetmanagement.sh
```

Local launch

```
source /opt/ros/<jazzy|humble|foxy>/setup.bash
source <ATOS_WORKSPACE>/install/setup.bash
Normal: ros2 launch atos launch_atosfleetmanagement.py
foxbridge:=False insecure:=True with_truck_simulator:=True
Off-path test: ros2 launch atos launch_atosfleetmanagement.py
foxbridge:=False insecure:=True with_truck_simulator:=True
simulator_2_lateral_offset_m:=10.0
All offsets: simulator_1_lateral_offset_m:=0.0
simulator_2_lateral_offset_m:=10.0 simulator_3_lateral_offset_m:=0.0
TruckObjectGUI: http://localhost:8420
```

Docker launch

```
docker compose -f docker-compose-fleetmanagement.yml up --build
WITH_TRUCK_SIMULATOR=True docker compose -f docker-compose-fleetmanagement.yml
up --build
WITH_TRUCK_SIMULATOR=False docker compose -f
docker-compose-fleetmanagement.yml up --build
```

Service / deployment

```
sudo cp scripts/atosfleetmanagement.env.example
/etc/default/atosfleetmanagement
sudo cp scripts/atosfleetmanagement.service
/etc/systemd/system/atosfleetmanagement.service
sudo systemctl daemon-reload
sudo systemctl enable --now atosfleetmanagement
```

Main runtime pieces

```
launch file:    ros2 launch atos launch_atosfleetmanagement.py
build script:   ./scripts/build_atosfleetmanagement.sh
run script:     ./scripts/run_atosfleetmanagement.sh
compose file:   docker-compose-fleetmanagement.yml
route asset:    conf/conf/RuralRoad_center_of_driving_lane_ccw.geojson
```

ROS interfaces

Send truck input to TruckObjectControl:

```
/atos/truck_objects/cot
Current temporary payload format:
id=<truck_id>;distance_m=<value>;tcp_connected=<0|1>
Read commanded speed from:
/atos/truck_objects/speed_command
Read live truck state for GUI/map from:
/atos/truck_objects/state
Useful inspection commands:
ros2 topic list
ros2 topic echo /atos/truck_objects/state
ros2 topic echo /atos/truck_objects/speed_command
```